

Analytic calculator for determination of γ -ray angular distribution coefficients and tensors in aligned and partially-aligned nuclei

A.M. Hurst^{a,b,*}, D.A. Matters^b, T. Kawano^c

^a*Department of Nuclear Engineering, University of California, Berkeley, California, 94720, USA*

^b*Nuclear Science Division, Lawrence Berkeley National Laboratory, Berkeley, California, 94720, USA*

^c*Theoretical Division, Los Alamos National Laboratory, Los Alamos, NM 87545, USA*

Abstract

A program has been developed to calculate a complete set of γ -ray angular distribution coefficients and statistical tensors in maximally- and partially-aligned nuclei. For practical nuclear structure and reaction purposes, there is no imposed constraint on any arguments that are likely to arise in the determination of these quantities. The program can also be used as a stand-alone vector-coupling calculator for the exact evaluation of Clebsch-Gordan and Racah coefficients, the closely-related Wigner 3- j , 6- j , and 9- j symbols, as well as Gaunt coefficients. These quantities, which frequently arise in quantum mechanical applications involving angular momentum coupling and recoupling schemes, provide the underlying machinery in angular distribution calculations.

Keywords: γ -ray angular distributions; statistical tensors; angular momentum; Clebsch-Gordan coefficients; Wigner coefficients; 3- j symbol; Racah coefficients; 6- j symbol; 9- j symbol; Gaunt coefficients; nuclear structure

PROGRAM SUMMARY

Program Title: PyGammaRAD: Python project for Gamma-Ray Angular Distributions

CPC Library link to program files: (to be added by Technical Editor)

*Corresponding author.

E-mail address: amhurst@berkeley.edu

Developer's repository link: <https://pypi.org/project/PyGammaRAD/> and <https://github.com/AaronMHurst/PyGammaRAD>

Code Ocean capsule: (to be added by Technical Editor)

Licensing provisions: MIT license

Programming language: Python

Nature of problem: Calculations based on angular momentum coupling schemes needed for the determination of γ -ray angular distribution coefficients and statistical tensors in maximally- and partially-aligned nuclei, in addition to other quantum mechanical applications that require the exact evaluation of angular momentum vector-coupling coefficients.

Solution method: Direct evaluation of sum formulae using conventional numerical computing methods to calculate angular momentum coupling coefficients and symbols, and the angular distribution coefficients and statistical tensors derived from these quantities.

1. Introduction

The angular distribution of γ rays emitted from oriented nuclear states carries rich spectroscopic information, encoding the multipolarities XL , where X denotes the electric (E) or magnetic (M) character and L the multipole order, of γ -ray transitions together with the associated spin J and parity π assignments of the nuclear levels involved. Many important parameters associated with angular distributions routinely used in nuclear physics data analysis are only available in limited tabular forms and there is a need to have libraries that support direct calculation to enable future automated analysis techniques and applications. However, before discussing the utility and formalism of γ -ray angular distributions, it is worthwhile to establish precise definitions, since the literature is not always unambiguous on this point.

A nuclear state of spin J possesses $2J + 1$ magnetic substates, each characterized by a quantum number m taking values from $-J$ to $+J$, defined with respect to a chosen quantization axis. When the populations of all magnetic substates are equal, the nucleus is said to be *unoriented*, and the resulting γ -ray angular distribution is perfectly isotropic, e.g., an unoriented $J = 2\hbar$ state has five associated magnetic substates each with equal population: $P(-2) = P(-1) = P(0) = P(+1) = P(+2) = \frac{1}{5}$. Any departure from this uniform population constitutes nuclear *orientation*. There are two classes of oriented nuclear states: *aligned* and *polarized*. Both classes of orientation

lead to anisotropies in the observed γ -ray angular distribution depending on the degree of alignment or polarization. A nucleus is said to be *aligned* if the population distribution is symmetric about $m = 0$, thus, our alignment condition is defined by $P(-m) = P(m)$. In the example of an aligned $J = 2\hbar$ state, this naturally implies $P(-2) = P(+2)$ and $P(-1) = P(+1)$, while the symmetry about $m = 0$ necessarily ensures that $P(|2|) < P(|1|) < P(0)$. In *polarized* nuclear states this symmetry is broken and $P(-m) \neq P(m)$. Consequently, for a polarized $J = 2\hbar$ state, to use the same example, this implies $P(-2) \neq P(+2)$ and $P(-1) \neq P(+1)$.

In reactions involving a beam of particles incident upon a target nucleus, such as compound-nucleus reactions [1], the beam direction naturally defines the quantization axis, often referred to as the z axis. Here, the reacting system possesses a reflection symmetry about the plane containing the beam axis for an unpolarized beam incident on an unpolarized target. This guarantees that the residual nuclear state is aligned but not polarized; a consequence of the axial-reflection symmetry of the reacting system [2]. Aligned states are further classified into two sub-classes depending on the degree of alignment: *maximum alignment* and *partial alignment*. When the alignment is modeled as a Gaussian distribution of magnetic-substate populations [3, 4] characterized by a width parameter σ , the degree of alignment is controlled by the ratio σ/J . In the limit of small σ/J (i.e., as $\sigma/J \rightarrow 0$) the Gaussian is narrow and population concentrates near $m = 0$, corresponding to the nuclear spin vector aligned predominantly perpendicular to the quantization axis — this is the *maximum* (or *complete*) *alignment* limit. As σ/J increases, population spreads symmetrically across the magnetic substates toward the $|m| = J$ extremes, and in the limit of large σ/J (i.e., as $\sigma/J \rightarrow \infty$) the distribution approaches the uniform statistical (isotropic) limit. *Partial alignment* refers to the intermediate regime in which the symmetry condition $P(m) = P(-m)$ is satisfied but population is attenuated and distributed across multiple substates with σ/J neither vanishingly small nor large.

The general form of the γ -ray *angular distribution* from an aligned nuclear state, measured as a function of the angle θ between the emitted γ -ray direction and the beam axis using a single detector, without any coincidence requirement, is given by

$$W(\theta) = \sum_{k=0,2,4,\dots} Q_k G_k(\tau) U_k B_k F_k P_k(\cos \theta), \quad (1)$$

where the sum runs over even values of k only. This expression should be dis-

tinguished from the γ - γ *angular correlation* function, in which θ denotes the angle between two successively detected γ rays in a coincidence measurement and the alignment of the intermediate state is established by detection of the first radiation [5, 6]; both take the same mathematical form, but the physical content of each factor differs. In Eq. (1), the F_k coefficients encode the nuclear structure information and are computed from appropriate Clebsch-Gordan and Racah coefficients following the formalism of Refs. [5, 7, 3], as described in Sect. 3. The B_k factors are the statistical population tensors characterizing the degree of alignment of the initial state; $B_0 = 1$ for all J (see Sect. 3.1.1) and $B_k = 0$ for all $k > 0$ in the unoriented statistical limit, i.e., for isotropic distributions. The Q_k factors are geometrical attenuation coefficients correcting for the finite solid angle subtended by the detector [8, 9]; a point detector would have $Q_k = 1$ for all k , while for a physical detector of finite size $Q_k < 1$ for $k > 0$, with the attenuation becoming more pronounced for higher-ranked terms [9]. Accurate determination of Q_k may be non-trivial depending on the experimental configuration. However, analytical approximations for various common source-detector geometries have been treated in a companion **Jupyter Notebook** and associated GitHub repository [10]. The $G_k(\tau)$ factors account for extranuclear perturbations based on hyperfine magnetic fields and electric quadrupole interactions with lattice field gradients that can disturb the nuclear spin orientation over the lifetime (τ) of the state, thereby attenuating the measured anisotropy [5, 11]. For short-lived states, however, these effects are negligible and $G_k \approx 1$ [12]. The U_k factors are deorientation coefficients that account for the depopulation of the aligned state through unobserved intermediate transitions in a cascade as discussed in Sect. 3.1.1. Finally, the $P_k(\cos \theta)$ terms are the Legendre polynomial functions of rank k . For oriented states only even-rank statistical tensors are non-zero, and the γ -ray angular distribution involves only even-order Legendre polynomials. This result follows from two independent constraints: the alignment condition itself eliminates odd-rank statistical tensors, and parity conservation in electromagnetic transitions independently suppresses odd- k terms [5]. It is also important to note that while γ -ray angular distributions are sensitive to the multipolarity order L and the spin sequence $J_i \rightarrow J_f$, they cannot distinguish between electric (EL) and magnetic (ML) radiation of the same multipole order [5, 11]; determination of the electromagnetic character requires additional measurements such as internal conversion coefficients or γ -ray linear polarization, e.g., Compton polarimetry [13].

In practice, following Yamazaki [3], the product $B_k F_k$ is frequently eval-

uated together and absorbed into a single anisotropy coefficient $A_k \equiv B_k F_k$, and Eq. (1) reduces in the idealized limit, assuming negligible deorientation for a short-lived state measured with a point detector such that $Q_k = G_k = U_k = 1$, to the compact form

$$W(\theta) = \sum_{k=0,2,4,\dots} A_k P_k(\cos \theta), \quad (2)$$

where the A_k anisotropy coefficients implicitly carry both the structural and alignment information of the initial state.

The theoretical framework underlying γ -ray angular distributions from aligned nuclei has a long history. The general theory of angular correlations of nuclear radiations was established by Hamilton [6] and developed into its modern form by Biedenharn and Rose [5] using the statistical tensor formalism of Fano [14] and the Racah coefficient machinery [15]. The extension to angular distributions from reaction-oriented states was developed by Litherland and Ferguson [2] and placed into a practically-convenient form by Krane, Steffen and Wheeler [8], who introduced a *directional-correlation-from-oriented-states* (DCO) functionality in terms of generalized F coefficients. The foundational experimental methodology was reviewed by Frauenfelder [11], while contemporary experiments continue to exploit the DCO ratio method, e.g., Ref. [16], enabling multipolarity characterization of emitted γ rays and J^π assignments of the levels involved. The DCO method is a type of angular correlation analysis based on γ - γ coincidence measurements where ratios of coincidence γ -ray intensities are measured at two different detector angles. In experimental applications involving arrays of detectors with large solid-angle coverage, such as Gammasphere, DCO ratios are usually deduced for two different subsets of angles: detectors arranged at forward/backward polar angles with respect to the beam direction (" $\theta_{F/B} \approx 180^\circ$ "), and those perpendicular to the beam (" $\theta_\perp \approx 90^\circ$ "). Here, the DCO ratio (see, e.g., Refs. [17, 18]) is deduced as

$$R_{\text{DCO}} = \frac{I_{\gamma_1}^{(\theta_{F/B})} [\text{Gate}_{\gamma_2}^{(\theta_\perp)}]}{I_{\gamma_1}^{(\theta_\perp)} [\text{Gate}_{\gamma_2}^{(\theta_{F/B})}]}, \quad (3)$$

where $I_{\gamma_1}^{(\theta_{F/B})} [\text{Gate}_{\gamma_2}^{(\theta_\perp)}]$ is the intensity of transition γ_1 in detectors at forward/backward angles $\theta_{F/B}$ when a coincidence-energy gate is set on transition γ_2 in detectors at perpendicular angles $\theta_\perp$, and similarly defined for

$I_{\gamma_1}^{(\theta_\perp)}[\text{Gate}_{\gamma_2}^{(\theta_{F/B})}]$. A closely related variant, sometimes referred to as an *angular-distribution-from-oriented-states* (ADO) ratio [19], measures the ratio of single-transition intensities at two different angles (or two different subsets of angles) without requiring a coincidence gate on a second γ ray:

$$R_{\text{ang}} = \frac{\Sigma_{\theta_{F/B}} I_\gamma}{\Sigma_{\theta_\perp} I_\gamma}. \quad (4)$$

This approach is frequently adopted in cases where transition intensities or counting statistics are low. In either case, the intensity ratios are calibrated against transitions of known multipolarity in well-studied nuclei providing a signature of the transition multipole order according to firmly-established limits for definitive $\Delta J = 1$ (dipole) or $\Delta J = 2$ (quadrupole) assignments. However, as noted by Piiparinen *et al.* [19], what is often loosely, though sometimes misleadingly, referred to as a DCO ratio in the literature is more accurately understood as an angular distribution from the reaction-oriented nuclear state. In this case, the alignment and angular anisotropy of γ -ray emission from the oriented state is exploited directly from the reaction dynamics rather than through a coincidence gate on a known transition. In practice, the two methods are implemented similarly using asymmetric γ - γ matrices in which one axis accepts events from all detectors regardless of angle and the other axis is incremented with events associated with a specific angular range, i.e., $\theta_{F/B} \approx 180^\circ$ or $\theta_\perp \approx 90^\circ$, and the distinction between them reduces to whether or not the all-angle axis is gated on a transition of known multipolarity. Angular-intensity ratios deduced in this manner have become routine with large-detector arrays in recent years, e.g., the TESSA spectrometer at the Daresbury Laboratory [20], the 8π array at Chalk River [21, 22], the HERA spectrometer at the Lawrence Berkeley National Laboratory (LBNL) [23], the EXOGAM array at the Grand Accélérateur National d'Ions Lourdes [24], the JUROGAM-II spectrometer at the University of Jyväskylä [25], the Euroball-IV array at the Institut de Recherches Subatomiques [26, 27], the Gammasphere array at LBNL [28, 29], and the Gammasphere array at the Argonne National Laboratory [30, 31, 32]. However, not all of these references adhere strictly to the coincidence-gated DCO definition of Eq. (3).

Heavy-ion fusion-evaporation reactions provide particularly favorable conditions for the production of strongly aligned residual nuclei [33, 34]. Projectile ions carry large orbital angular momenta into the compound system, leaving the angular momentum vector of the residual nucleus strongly

aligned perpendicular to the beam axis following evaporation of a few neutrons. The neutron evaporation process generates an approximately Gaussian distribution of magnetic substates [34], and the resulting pronounced γ -ray angular distribution anisotropies make these measurements a powerful spectroscopic tool for determining transition multipolarities and nuclear level spins and parities. This near-maximal alignment regime continues to be exploited in contemporary experimental programs, e.g., as demonstrated in low-energy inelastic proton scattering experiments using GRETINA [13] where the strongly aligned nuclear states produced enable determination of transition multipolarities and electromagnetic character.

For reactions involving lighter projectiles at lower incident energies, such as inelastic neutron-scattering measurements [35, 36], the degree of alignment achieved is substantially less and a partial-alignment treatment is required. In such cases the Gaussian parameterization provides a practical and experimentally justifiable framework [3, 4], and γ -ray angular distribution measurements continue to provide important spectroscopic information [37, 38, 16]. The same statistical tensor formalism has recently been applied in the context of nuclear fission, where Monte Carlo calculations have been used to quantify the spin alignment of fission fragments following neutron emission [39].

The resources most widely used in the analysis of experimental data have long been the *Tables of Coefficients for Angular Distribution of Gamma Rays from Aligned Nuclei* by Yamazaki [3], the *Angular Distributions of Gamma Rays in Terms of Phase-Defined Reduced Matrix Elements* by Rose and Brink [7], and the companion partial-alignment tables of Der Mateosian and Sunyar [4]. There are, however, significant practical limitations in the use of these tabulated resources. The tables of Yamazaki [3] provide the population tensor for ranks $k = 2, 4$, and 6 , and the angular distribution coefficients for tensor ranks $k = 2$ and 4 only, covering multipole orders $1 \leq L \leq 3$ associated with initial integral spins $1 \leq J \leq 15 \hbar$ and half-integral spins $3/2 \leq J \leq 31/2 \hbar$. Rose and Brink [7] extend the multipole coverage through to $L = 4$ and the tensor rank to $k = 2, 4, 6$, and 8 , but impose their own spin restrictions depending on the quantity in question: the angular distribution coefficients are tabulated up to highest initial spins of $J = 10 \hbar$ (integral) and $J = 19/2 \hbar$ (half-integral); the deorientation coefficients are given for multipole orders up to $L = 5$; the statistical tensor is tabulated for a maximum of seven magnetic substates, reaching spins of $J = 12 \hbar$ (integral) and $J = 23/2 \hbar$ (half-integral); and the population tensor is restricted to a maximum or-

bital angular momentum difference of $\Delta l = 4$, with reaction-channel spin $0 \leq s \leq 4$ and target spin $0 \leq J \leq 5\hbar$ for integral spins, or channel spin $1/2 \leq s \leq 7/2$ and target spin up to $1/2 \leq J \leq 9/2\hbar$ for half-integral spins. The partial-alignment tables of Der Mateosian and Sunyar [4] tabulate partial-alignment attenuation-anisotropy coefficients for tensor ranks $k = 2$ and $k = 4$ only, evaluated at a discrete set of Gaussian width parameters within the range $0.1 \leq \sigma/J \leq 2.0$, spanning integral spins over $1 \leq J \leq 26\hbar$ and half-integral spins from $3/2 \leq J \leq 51/2\hbar$, and restricted to dipole and quadrupole transitions only. Fusion-evaporation reactions at modern stable ion-beam facilities routinely populate rotational cascades well beyond these ranges, rendering the tables inadequate in many contemporary cases. Moreover, these resources exist only in printed form, making their use inconvenient, potentially error prone, and no accompanying codebase was released alongside.

Several programs exist that address subsets of the functionality provided by the current project PyGammaRAD, including a lineage of increasingly capable Fortran implementations for the underlying vector-coupling coefficients [40, 41, 42, 43, 44], computer algebra system approaches based on proprietary software [45, 46], as well as more recent Java [47, 48] and symbolic Python-based [49] open tools for the calculation of the coupling coefficients alone. Macias *et al.* [50] developed a Fortran program specifically for calculating γ - γ directional correlation coefficients and mixing ratios. This work was later modified by Rouse *et al.* [51] to include analysis of angular distributions for γ rays of mixed multipolarities from partially-aligned nuclei. However, these codes [50, 51] are written in a language that is no longer straightforwardly accessible to many users and can only be used to calculate a limited subset of the angular distribution coefficients presented in the adopted literature [3, 4, 7]. Furthermore, none of the aforementioned capabilities integrates the vector-coupling machinery with the full apparatus of γ -ray angular distributions described in the foundational tabular works of Yamazaki [3], Rose and Brink [7], and Der Mateosian and Sunyar [4], in a single unified openly available package. PyGammaRAD addresses this gap: an open-source toolkit written in Python, it performs standard double-precision floating-point arithmetic and extended-precision arithmetic, as needed, that consolidates Clebsch-Gordan, Racah, Wigner 3- j , 6- j , 9- j , and Gaunt coefficient calculations, together with the complete set of maximum [3, 7] and partial [3, 4] alignment angular distribution formalism presented in the associated canonical reference material. To that end, PyGammaRAD is a self-contained tool based on conventional nu-

merical computing that is portable, extensible, and accessible to the broad community of users familiar with the Python programming language and its associated libraries, making it straightforward to integrate into larger code-bases. The program reproduces all tabulated data in Refs. [3, 4, 7] with full numerical precision — in doing so, a small number of typographical errors in Ref. [7] were identified — and can readily evaluate angular-distribution coefficients for nuclei populated to arbitrarily high spin over a much wider range of multiplicities and tensor ranks than any of the original works.

A brief overview of the program structure is described in Sect. 2. The formalism adopted for the calculation of γ -ray angular distribution coefficients is described in Sect. 3. Section 4 provides a discussion of the benchmarking and validation procedures. In Sect. 5 a summary and outlook is presented. Finally, an overview of the underlying vector-coupling formalism commonly used in quantum physics applications based on the theory of angular momentum is presented in the Appendix A.

2. Structure of the program

A Python implementation of a suite of calculators has been written to return precise analytic results for the angular distribution coefficients and tensors defined in Sect. 3, together with the associated angular momentum coupling coefficients which are required in the determination of these quantities. The closely related angular momentum coupling symbols are also defined and coded into the suite, thus the complete software package offers broader application beyond the scope of the angular distribution focus of this manuscript. The program is structured in a modular fashion and a single source file `am_formulae.py` contains separate classes based on multiple inheritance for the angular momentum coupling coefficients: `am_formulae.ClebschGordan`, `am_formulae.Wigner3j`, `am_formulae.Racah`, and `am_formulae.Wigner9j` for the Clebsch-Gordan coefficient, Wigner-3j symbol, Racah coefficient, and Wigner-9j symbol calculators, respectively. The Wigner-6j symbol is strongly related to the Racah coefficient through a simple phase factor (see Appendix A.5) and, therefore, a member function for its evaluation is also contained within the same `am_formulae.Racah` class. The calculation of the Wigner-9j symbol is also simplified using `am_formulae.Racah` class-inherited methods, as described in Appendix A.6. Gaunt coefficients can be decomposed into a product of two Wigner-3j symbols (see Appendix A.3) and can, therefore, be evaluated within the `am_formulae.Wigner3j` class. The

callable methods used to return the exact solution of these vector coupling coefficients are then contained within `AngularMomentumCalculations` class of the `am_methods.py` module.

The source file `angular_distributions.py` contains the appropriate member functions of the `AngularDistributions` class needed to calculate the γ -ray angular distribution coefficients and tensors listed in the tabulated data of Yamazaki [3], as well as those in the tables of the appendix in the review article by Rose and Brink [7], and the attenuated coefficients tabulated by Der Mateosian and Sunyar [4]. To aid in the determination of the overall angular distribution calculation, the `angular_distributions.Legendre` class contains the first 11 polynomials of the Legendre series coded in terms of $\cos\theta$. Finally, the `tables.py` module contains classes and methods for handling and manipulating the tabulated data of Refs. [3, 4, 7]. Effectively, this module serves as an API for interacting with the published tables in these works and also allows users to fully regenerate the tables in convenient machine-readable CSV and JSON formats.

All of these classes are consolidated through multiple inheritance into a single top-level `AngularMomentum` class, which serves as the unified user-facing API to the complete PyGammaRAD library. Consequently, once an instance of `AngularMomentum` is created, all methods — regardless of which source module they originate from — are directly accessible through that single instance, as illustrated in the next Sect. 2.1

2.1. Running the program

The PyGammaRAD software package is open source, freely available on GitHub [52] and the Python Package Index (PyPI) repository [53], and comes packaged with a permissive MIT license. Building and installation are straightforward and are detailed in the accompanying `README` documentation on the GitHub and PyPI landing pages. The calculation of all quantities and utilization of associated functions is enabled by creating an instance of the `AngularMomentum` class after importing the PyGammaRAD library, for example:

```

1  In [1]: import PyGammaRAD as pg
2  In [2]: am = pg.AngularMomentum()
3
4  # Calculate Clebsch-Gordan coefficient <j1=5/2 m1=3/2
   j2=5/2 m2=-1/2 | j=1 m=1>
5  In [3]: am.cg(2.5, 1.5, 2.5, -0.5, 1, 1)
6  Out[3]: -0.4780914437337574 # -2*sqrt(2/35)

```

```

7
8 # Calculate coefficient F(k=2,Jf=0,L1=2,L2=2, Ji=2)
9 In [4]: am.calc_F(2, 0, 2, 2, 2)
10 Out[4]: -0.5976143046671966
11
12 # Compare to result in "Table 2(a)" [Yamazaki 1967]
13 In [5]: am.get_row_table2(2,0,2,2,2,coeff='F')
14 Out[5]: -0.59761
15
16 # Print "Table 2(a)" from Ref. [Yamazaki 1967] to file
   in JSON format
17 In [6]: am.yamazaki2file('T2A', 'JSON')
18 INFO:PyGammaRAD.log_handlers:YamazakiTable2a.json

```

To help illustrate various workflows and the overall utility of the software, including verification of methods against published results (see Sect. 4), PyGammaRAD comes shipped with several different Jupyter Notebooks available from the project GitHub repository [52]. All PyGammaRAD classes and member functions have supporting docstrings that can be invoked at the console as described in the README. These docstrings provide a short explanation of the function, any arguments passed, the return value, as well as exceptions that are raised by the function. Each docstring is also presented with at least one known working example to showcase the corresponding method execution. All angular distribution and angular momentum coupling methods available to the PyGammaRAD library are also summarized in the associated README documentation, along with methods enabling retrieval and manipulation of the tabulated data in Refs. [3, 4, 7].

2.2. Logging, testing, and dependencies

A logging framework has been implemented within PyGammaRAD whereby messages are categorized appropriately using different levels of severity. By default, logs are only directed to the console; however, an application log can also be redirected to file upon request. The implemented logging system provides a centralized and standardized way to handle the field reporting of application events such as errors and warnings. The logging system is configured to provide useful information and metadata regarding such events, including timestamps, module names, function names, and line numbers, thus enabling a richer context for debugging and analysis, making it easier for the user to monitor and troubleshoot issues.

A suite of Python modules comprising more than 200 unit tests has been developed as part of this project to help validate and benchmark its performance against published results, verification of return types, and raising of exceptions when certain conditions arise. All angular distribution coefficients and tensors presented in Sect. 3, together with the angular momentum coupling coefficients presented in Appendix A, have dedicated modules that comprise the test suite. The test scripts are run to ensure compliance with the local Python environment and have been successfully tested across multiple Python versions 3.6 to 3.13 inclusive on both Linux and Mac OS platforms. The PyGammaRAD software package has the following third-party Python-package dependencies: `numpy`, `pandas`, and `pytest`. There is also an additional `scipy` package dependency to run certain actions in the accompanying notebooks.

3. Formalism used in the calculations

The calculators used in the evaluation of the overall angular distribution coefficients and tensors are based on the well-known sum formulae for the coupling of angular momenta vectors which have been extensively described in various dedicated texts on the subject matter, e.g., Refs. [54, 55, 56, 57], as well as in the published literature, e.g., Refs. [40, 41, 42, 43, 44, 45, 58, 59, 47, 60]. The vector-coupling routines are described in Appendix A, while the quantities explicitly of relevance to angular distributions are described in the following subsections.

3.1. γ -ray angular-distribution coefficient calculator

Methods have been developed based on the formalism outlined by Yamazaki [3] and Rose and Brink [7] to calculate angular distribution functions in maximally-aligned nuclei. Additionally, we have further developed methods for partial alignment in nuclei based on the formalism outlined by Der Mateosian and Sunyar [4], which in itself is an extension of the earlier work by Yamazaki [3]. Here, we present an overview of the corresponding expressions used to calculate the various functions and coefficients that are needed to reproduce all the tabulated data in Refs. [3, 4, 7]. Furthermore, these methods can be applied for values of J , L , m , and k well beyond the ranges presented in the original works of Yamazaki [3], Rose and Brink [7], and Der Mateosian and Sunyar [4].

Two important special cases are worth noting. Firstly, for states characterized by $J = 0$ the single magnetic substate $m = 0$ means no preferred direction can be established and emission is necessarily isotropic. Secondly, and for states with $J = 1/2 \hbar$, the two magnetic substates $m = \pm 1/2$ are equally populated under reaction-symmetry conditions imposed by the reflection symmetry about the beam axis, forcing a symmetric distribution through the $m = \pm 1/2$ substates due to the orbital angular momentum of the incoming projectile, which again gives rise to an isotropic distribution. Accordingly, in the canonical reference material [3, 4, 7], there are no initial states with $J_i = 0$ or $J_i = 1/2 \hbar$ since the associated anisotropy coefficients are expected to vanish in these cases.

The adopted formalism based on complete and partial alignment are discussed, in turn, below.

3.1.1. Maximum nuclear alignment

The overall angular distribution function [3] for a γ -ray transition from an initial state to a final state $J_i \rightarrow J_f$ may be adequately expressed using the leading-order terms from Eq. (2), thus

$$W(\theta) = 1 + A_2 P_2(\cos \theta) + A_4 P_4(\cos \theta), \quad (5)$$

where A_k are the anisotropy coefficients and P_k are the Legendre polynomials of order k . In principle, the series extends to higher k as per Eqs. (1) and (2) earlier, however, the truncation at $k = 4$ is conventional for the cases addressed by the original tables [3] since higher-rank terms vanish for low spins or are negligible. We should also note that PyGammaRAD is not limited in this way and distributions can be straightforwardly calculated for higher-rank $k > 4$ terms.

To obtain A_k in Eq. (5), we first need to calculate the statistical population tensor B_k and the F_k angular distribution coefficient [3]. In the case of complete nuclear alignment, we can write the statistical population tensor for integral and half-integral spin, respectively, as

$$\begin{aligned} B_k(J) &= (-1)^J \sqrt{2J+1} \langle J0J0|k0 \rangle; \\ B_k(J) &= (-1)^{J-\frac{1}{2}} \sqrt{2J+1} \langle J\frac{1}{2}J-\frac{1}{2}|k0 \rangle, \end{aligned} \quad (6)$$

where $\langle J0J0|k0 \rangle$ and $\langle J\frac{1}{2}J-\frac{1}{2}|k0 \rangle$ are Clebsch-Gordan coefficients for given

J and k . The F_k angular distribution coefficient is given as

$$F_k(J_f L_1 L_2 J_i) = (-1)^{J_f - J_i - 1} \sqrt{(2L_1 + 1)(2L_2 + 1)(2J_i + 1)} \\ \times \langle L_1 1 L_2 - 1 | k 0 \rangle W(J_i J_i L_1 L_2; k J_f), \quad (7)$$

where $W(J_i J_i L_1 L_2; k J_f)$ is a Racah coefficient, and L_1 and L_2 are the angular momentum multipole orders for a mixed transition whereupon the lowest two orders are considered such that the interfering multipole $L_2 = L_1 + 1$. In the case of a pure transition $L_1 = L_2 = L$. For the ideal case of maximal alignment we can then compute A_k using

$$A_k^{\max}(J_i L_1 L_2 J_f) = \frac{1}{1 + \delta_\gamma^2} [B_k(J_i) F_k(J_f L_1 L_1 J_i) + 2\delta_\gamma B_k(J_i) F_k(J_f L_1 L_2 J_i) \\ + \delta_\gamma^2 B_k(J_i) F_k(J_f L_2 L_2 J_i)], \quad (8)$$

where δ_γ is the γ -ray mixing ratio [61]. The mixing ratio can be defined in terms of the reduced matrix elements of the associated electromagnetic multipole operators:

$$\delta_\gamma = \frac{\langle J_f || \mathcal{M}(L_2) || J_i \rangle}{\langle J_f || \mathcal{M}(L_1) || J_i \rangle}. \quad (9)$$

In the case of a pure transition when $L_1 = L_2$ there is no mixing and $\delta_\gamma = 0$. For other cases where $|\delta_\gamma| > 0$, δ_γ should be obtained from experiment, e.g., [16, 62], and provided as an input parameter to PyGammaRAD methods as appropriate. A survey of $M1 + E2$ mixing ratios from experimental angular distribution and correlation data is described and the results are tabulated in Ref. [61]. Alder and Steffen [63] provide a comprehensive discussion of electromagnetic multipole operators and their associated matrix-element transformation properties.

The anisotropy coefficient, expressed in Eq. (7) above, can alternatively be described in terms of the closely related R_k angular distribution coefficient [7]:

$$R_k(L_1 L_2 J_i J_f) = (-1)^{1 + J_i - J_f + L_2 - L_1 - k} \sqrt{(2L_1 + 1)(2L_2 + 1)(2J_i + 1)} \\ \times \langle L_1 1 L_2 - 1 | k 0 \rangle W(J_i J_i L_1 L_2; k J_f). \quad (10)$$

Clearly, Eqs. (7) for F_k and (10) for R_k differ only in their phases and can be related through the following expression

$$R_k(L_1 L_2 J_i J_f) = (-1)^{L_1 - L_2 + k} F_k(J_f L_1 L_2 J_i), \quad (11)$$

and so the anisotropy coefficient given in Eq. (8) may then be recast by replacing the F_k coefficients with their corresponding R_k coefficients [7], thus allowing for the overall distribution function [Eq. (5)] to be expressed in terms of R_k instead.

In a γ -ray cascade, for a state with spin J_f whose formation is dependent entirely upon the preceding $J_i \rightarrow J_f$ transition, it is useful to introduce the U_k angular distribution coefficient [3], often referred to as the deorientation (or depolarization) coefficient [64], which describes the effect of unobserved transitions preceding the observed radiation. This coefficient has the effect to alter the magnetic substate populations thereby reducing the degree of orientation ($U_k < 1$). It is dependent upon the multipolarities of the unobserved transitions, together with the angular momenta of the intermediary states, and may be written as

$$U_k(J_i L_1 L_2 J_f) = \frac{1}{1 + \delta_\gamma^2} [u_k(J_i L_1 J_f) + \delta_\gamma^2 u_k(J_i L_2 J_f)], \quad (12)$$

where

$$u_k(J_i L_1 J_f) = (-1)^{J_i + J_f - L_1} \sqrt{(2J_i + 1)(2J_f + 1)} W(J_i J_i J_f J_f; k L_1). \quad (13)$$

Alternatively, Eq. (13) above may also be expressed as a ratio of Racah coefficients [7] given by

$$\begin{aligned} u_k(J_i L_1 J_f) &= (-1)^k \frac{W(J_i J_f J_i J_f; L_1 k)}{W(J_i J_f J_i J_f; L_1 0)} \\ &= (-1)^k \frac{W(J_i J_i J_f J_f; k L_1)}{W(J_i J_i J_f J_f; 0 L_1)}. \end{aligned} \quad (14)$$

If we consider the two-step γ -ray cascade involving transitions between states from $J_1 \rightarrow J_2 \rightarrow J_3$ as shown in Fig. 1a, where J_1 is the initially populated aligned state satisfying the alignment condition $P_{m_i}(J_1) = P_{-m_i}(J_1)$, with the normalization condition $\sum_{m_i=-J_1}^{+J_1} P_{m_i}(J_1) = 1$ satisfied, and characterized by statistical tensors $B_k(J_1) \neq 0$ for even $k > 0$, the distribution formula expressed in Eq. (5) will need to be modified for the second γ ray, labeled γ_{23} in Fig. 1a, which then has the following distribution [7]

$$W(\theta) = \sum_{k=0,2,4} U_k(J_i = J_1, J_f = J_2) A_k(J_i = J_2, J_f = J_3) P_k(\cos \theta). \quad (15)$$

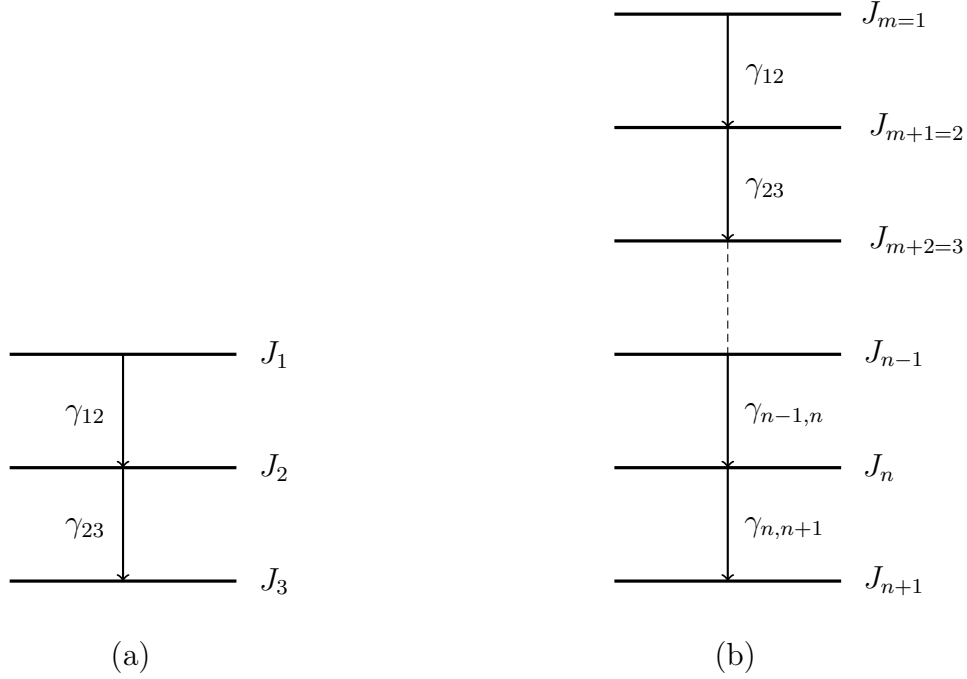


Figure 1: (a) Two-step γ -ray cascade. (b) Multi-step γ -ray cascade.

Equation (15) can be generalized for the n^{th} γ ray in a multi-step cascade, as depicted in Fig. 1b, by including a product of U_k terms for each preceding non-observed γ ray, i.e., all γ rays other than $\gamma_{n,n+1}$ shown in Fig. 1b, such that

$$\begin{aligned}
 W(\theta) = & \sum_{k=0,2,4} \left[A_k(J_i = J_n, J_f = J_{n+1}) P_k(\cos \theta) \right. \\
 & \times \left. \prod_{\substack{J_{n-1}, J_n \\ J_m, J_{m+1}}} U_k(J_i = J_m, J_f = J_{m+1}) \right]. \quad (16)
 \end{aligned}$$

Another important quantity that arises in the treatment of γ -ray angular distributions is the statistical tensor. Rose and Brink [7] define this tensor in terms of both spin J and its quantized magnetic substate projection m , thus

$$\rho_k(J, m) = (2 - \delta_{m,0}) \sqrt{2J+1} (-1)^{J-m} \langle JmJ - m | k0 \rangle, \quad (17)$$

this equation, therefore, represents the tensor element for a definite magnetic substate m without population weighting. Here, the Kronecker delta

function $\delta_{m,0} = 1$ for $m = 0$; otherwise $\delta_{m,0} = 0$ for all other values of m . This expression removes the dependency of any knowledge of the associated population parameters [3], and accordingly, $\rho_k(J, m)$ can then be calculated for pure individual m -substate projections for a given J .

The final coefficient that we consider in this section is the S_k coefficient [7] which characterizes the alignment of a nuclear state J when that state is formed in a particle-induced reaction and is written as

$$S_k(l_1 l_2 J s) = (-1)^{s-J} \sqrt{(2l_1 + 1)(2l_2 + 1)(2J + 1)} \\ \times \langle l_1 0 l_2 0 | k 0 \rangle W(J l_1 l_2; k s). \quad (18)$$

Here l_1 and l_2 are the orbital angular momenta of the incident particle relative to the target nucleus — two interfering partial waves representing distinct ways the incident particle can couple its orbital motion to form the same compound-nucleus state J — and s denotes the reaction-channel spin which corresponds to the vector sum of the spin of the incident particle and the spin of the target nucleus. All three entrance-channel quantities, i.e., l_1 , l_2 , and s , therefore pertain to the incident particle and target nucleus, rather than the exit channel of the reaction. The S_k coefficient encodes the dependence of the alignment coefficients $B_k(J)$ on this partial-wave content (note that S_k is an input to $B_k(J)$ [7], not a replacement for it; both are referred to as population tensors in the literature but they are distinct quantities). This formalism applies to resonant capture reactions such as (p, γ) , (n, γ) , and (α, γ) , as well as to inelastic-scattering reactions with significant compound-nucleus contributions, such as (p, p') and (n, n') . In this respect S_k is distinct in character from R_k and U_k : rather than characterizing the observed γ -ray transition or an unobserved cascade step, it describes the reaction mechanism responsible for establishing the initial nuclear alignment.

3.1.2. Partial nuclear alignment

In real experimental scenarios, the degree of alignment is often only partial [16, 38, 37] and modifications to the overall distribution function given by Eq. (5) need to be accounted for. Here, an attenuation coefficient is defined such that

$$\alpha_k(J) = \frac{\rho_k(J)}{B_k(J)}, \quad (19)$$

where ρ_k represents the statistical tensor [3] for a state J defined by the population parameters P_m that specify the formation process for this state

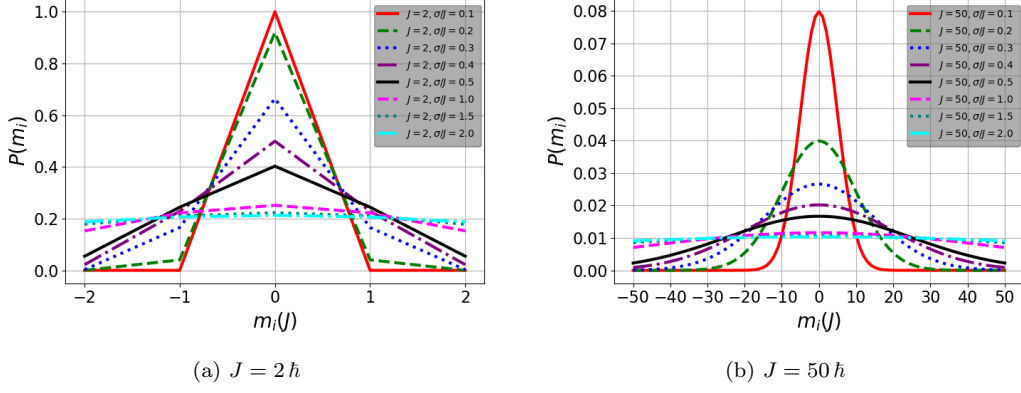


Figure 2: Magnetic substate populations $P(m_i)$ as a function of $m_i(J)$, shown for aligned (a) $J = 2 \hbar$ and (b) $J = 50 \hbar$ states. In each case, the Gaussian-width control parameter is varied from $\sigma/J = 0.1$, where the distributions approach maximal alignment with $P(m_i)$ concentrated on $m = 0$, to $\sigma/J = 2.0$, where the distributions approach an unoriented uniform distribution for all m .

and is given by

$$\rho_k(J) = \sqrt{2J+1} \sum_{m=-J}^J (-1)^{J-m} \langle JmJ - m | k0 \rangle P_m(J). \quad (20)$$

The expression embodied by Eq. (20) represents the full-weighted statistical tensor and differs from the earlier Eq. (17), which describes the statistical tensor as a function of J and m individually, in that this expression is summed over all values of m weighted by their corresponding population parameter. In cases where the alignment is only partial, the states are oriented with attenuated magnetic-substate populations although the symmetry about $m = 0$ still holds, i.e., $P_m(J) = P_{-m}(J)$, and $\rho_k(J)$ vanishes for odd- k . If $P_m(J)$ is unknown, then $\rho_k(J)$ given by Eq. (20) in the work of Yamazaki [3] is, in general, difficult to evaluate. Experimentally, however, it is often justifiable to assume a Gaussian distribution of magnetic substates [3, 4, 37] characterized by the half-width parameter σ such that the m -substate population parameter may be calculated in accordance with the following expression

$$P_m(J) = \frac{\exp\left(-\frac{m^2}{2\sigma^2}\right)}{\sum_{m=-J}^J \exp\left(-\frac{m^2}{2\sigma^2}\right)}. \quad (21)$$

Under this assumption, the dependence of the partial-alignment attenuation coefficient $\alpha_k(J)$ is controlled by the Gaussian-width distribution parameter σ/J , and the modified angular distribution function may then be written as

$$W(\theta) = 1 + \alpha_2 A_2 P_2(\cos \theta) + \alpha_4 A_4 P_4(\cos \theta). \quad (22)$$

Again, the explicit sum over the $k = 0, 2, 4$ terms reflects the practical truncation imposed in the original tables [4]; however, **PyGammaRAD** is not restricted to $k \leq 4$. The $\alpha_k A_k$ term represents the attenuated anisotropy coefficient that can be compared with experimental anisotropy coefficients, frequently labeled a_k [16, 38, 37]. Furthermore, in the analysis of such experimental data, the multipole mixing ratio δ_γ — defined earlier by Eq. (9) in Sect. 3.1.1 — is typically extracted by comparing the measured ratio a_2/a_4 of angular distribution coefficients with theoretically-calculated values for a given σ/J and interpolating to find the value of δ_γ consistent with the experimental ratio [4, 16].

For illustrative purposes, in Fig. 2 we show the transition from near-complete alignment in $J = 2\hbar$ and $J = 50\hbar$ states centered on $m = 0$ ($\sigma/J = 0.1$), with vanishingly small populations in the corresponding $|m| > 0$ substates, to a near purely-statistical unoriented distribution revealing approximately uniform populations across all m ($\sigma/J = 2.0$). The attenuation in alignment as σ/J increases is also clear in this figure; various degrees of partial alignment in the intermediate region between these two limits are also shown.

4. Verification of the programs

The methods presented in Sect. 3.1 were checked against the tabulated angular distribution coefficients in the data tables of Yamazaki [3], Rose and Brink [7], and Der Mateosian and Sunyar [4]. In all cases, other than a few typographical errors that were found in the R_k and ρ_k tables of Rose and Brink [7],¹ the published data are reproduced precisely, which further

¹Typographical errors discovered in the angular distribution R_k [Eq. (10)] and statistical tensor ρ_k [Eq. (17)] coefficients tabulated in the appendix of Rose and Brink [7] by comparison with **PyGammaRAD** [52, 53].

Coefficient	Rose and Brink [7]	PyGammaRAD [52, 53]
$R_{k=6}(L_1 = 3 L_2 = 4 J_i = 3 J_f = 6)$	-0.0663	-0.0063
$R_{k=2}(L_1 = 2 L_2 = 2 J_i = 17/2 J_f = 13/2)$	0.3776	-0.3776
$\rho_{k=2}(J = 4 m = 0)$	-1.1366	-1.1396

serves as an implicit verification of the vector-coupling calculation machinery underpinning the evaluation of the Clebsch-Gordan and Racah coefficients that are needed in the overall determination of the various calculated angular-distribution coefficients and statistical tensors. As a further check, the Clebsch-Gordan coefficients and Wigner 3- j , 6- j , and 9- j symbols, based on the formalism outlined in Appendix A, were checked against all tabulated values published by Stevenson [47]. Complete agreement with the tabulated values of Stevenson [47] was obtained with the exception of a single Wigner-9 j entry; the affected entry appears to be a typographical error which is detailed in the supplementary **Jupyter Notebook** [52] and no correction to the underlying formalism [47, 48] is implied. We also checked the calculation of Wigner-9 j symbols with **Mathematica** [46] as well as tabulated results published by Tamura [40] and Wei [58]. Clebsch-Gordan and Racah coefficients tabulated by Tamura [40] were also confirmed using our calculator. Additionally, we were able to confirm the calculation of the angular momentum couplings against the online Wigner coefficient calculator based on the root-rational-fraction program developed by Stone [44]. Further tests were also performed and confirmed against the **SymPy** symbolic mathematics library [49], which provides a complete suite of vector-coupling functions, including Clebsch-Gordan, Racah, Wigner 3- j , 6- j , 9- j , and Gaunt coefficients, within a modern open Python environment. The published literature on tabulated Gaunt coefficients is more limited; although we were generally able to reproduce the results reported by Yükcü [65]. However, a number of errors were found in the proposed couplings of this work due to a failure to satisfy the condition $\sum_{m_i=1}^{m_i=3} m_i = 0$. These coupling errors are addressed in the accompanying **Jupyter Notebook** [52] and we have confirmed our results by comparison with **SymPy** [49].

The companion **Jupyter Notebooks** distributed with this project [52] also contain actions confirming all the aforementioned published results [3, 4, 7, 47, 44, 40, 58, 65], thus providing the user with convenient reference workflows and a means of inspection to perform independent checks of the data and calculation methods. Furthermore, as part of the test-driven development process, an extensive suite of unit tests has also been written. These unit tests were, in part, developed to corroborate the well-established results of Refs. [3, 4, 7, 47, 44, 40, 58, 65]. Unit tests have been written to establish the correct behavior for the complete apparatus of angular-distribution formalism presented in Sect. 3.1 as well as the angular momentum formulae of Appendix A upon which vector-coupling suite of **PyGammaRAD** is based.

It is intended for users to run through the test script to demonstrate the established conformity of all results within their local environment.

We can also assert that Eq. (13) [3] and Eq. (14) [7] are equivalent formal methods for determining the u_k angular distribution coefficient. This claim was checked by performing an extensive set of unit tests over all integral and half-integral results presented in the data tables of Yamazaki [3]. These tests are also distributed as part of the test suite.

Angular momentum couplings based on large integer arguments have also been tested. `PyGammaRAD` reproduces all very large angular momenta couplings reported by Stevenson [47] as well as those by Johansson and Forssén [60] – with the exception of the most extreme cases involving $j = 10,000$ or $j = 50,000$ [60], for which `PyGammaRAD` is not a suitable tool. The built-in integer type of the Python programming language automatically handles arbitrarily large integers, so there is no inherent size limit beyond the available memory. However, for certain very large integers, for example, factorials of large numbers, the native `math.sqrt` method may not execute the conversion to a floating-point representation, resulting in an `OverflowError` exception. To handle such exceptions, `PyGammaRAD` employs a dual-path arithmetic strategy. For the Clebsch-Gordan coefficient, a primary path first attempts evaluation using standard IEEE 754 double-precision floating-point arithmetic throughout, computing both the square-root prefactor and the alternating summation (see Eq. (A.1), Appendix A.1) in native Python floats. If an `OverflowError` is raised at any point, the entire calculation is automatically retried using the Python method `decimal.Decimal` with extended-precision arithmetic at 1000 significant digits. In this fallback approach, the prefactor is evaluated via `decimal.Decimal.sqrt` and the alternating sum gets accumulated in a `Decimal` accumulator before conversion back to a float result upon return. Wigner 3- j symbols (see Eq. (A.5), Appendix A.2) and Gaunt coefficients (see Eq. (A.7), Appendix A.3) are closely related to the Clebsch-Gordan coefficient, allowing for the same inherited treatment via standard double floating-point or extended precision, as appropriate. For the Racah coefficient, which involves a product of four triangular delta factors each containing multiple factorials together with a similarly structured alternating sum (see Eq. (A.8), Appendix A.4), the potential for numerical overflow is more acute; accordingly, `Decimal` arithmetic is employed unconditionally rather than as a fallback, ensuring that no precision is lost in intermediate calculations regardless of the magnitude of the angular momentum arguments. Since the Wigner 6- j symbol is obtained directly from the Racah

coefficient through a simple phase factor (see Eq. (A.13), Appendix A.5), and the Wigner 9- j symbol is decomposed into a sum of triple products of Wigner 6- j symbols (see Eqs. (A.17) and (A.18), Appendix A.6), both quantities inherit the same extended-precision treatment. It is worth noting that our tests of the Wigner 9- j method encompass complete parity coverage: integral and half-integral cases, including mixed and large angular momentum entries, have been verified against published results and confirmed in the accompanying **Jupyter Notebooks**.

During development, various strategies were considered to avoid the pitfalls of overflow and underflow. Although the methods described below are not actively used to return the results of any vector-coupling calculations described here, they have been left in the code for potential future use. The triangular delta factor (see Eq. (A.2), Appendix A.1 and also Appendix A.4) may be obtained independently via three different methods available within the `am_formulae.Racah` class: direct evaluation `tri_factor`; exponentiated logarithm `tri_factor_exp`; and logarithmic evaluation (to defer exponentiation) `tri_factor_log`. The validity of the triangular delta-factor methods have been well-established through unit tests whereupon they were incorporated into the vector-coupling calculations described in Appendix A during earlier phases of development. An integer square root method based on Newton’s iterative method [66], implemented as a member function of the `am_formulae.Newton` class as `isqrt`, is also provided as a utility for handling square roots of very large integers through refinement of an initial estimate based on the argument bit length until convergence with the correct integer square root is achieved, and has been tested against the native `math.sqrt` method using smaller integer arguments to ensure reproducibility. Notebook actions and unit tests have also been developed to demonstrate the validity of the `am_formulae.Newton.isqrt` method. Finally, although we adopt the native `math.factorial` method and ensure arguments are interpreted as integers, we also performed checks against a user-defined recursive factorial method in addition to a Γ function where $\Gamma(n + 1) = n!$. Consistency has also been demonstrated for these three methods, all of which are members of the `am_formulae.Factorial` class.

5. Summary

A set of calculators has been developed in Python to return precise γ -ray angular-distribution coefficients and statistical tensors that are useful in

the interpretation of nuclear structure data. These calculators have been benchmarked against the original canonical works of Yamazaki [3] and Rose and Brink [7] for maximally-aligned nuclei, as well the tables of attenuation coefficients published by Der Mateosian and Sunyar [4] for partially-aligned nuclei. Although the range of arguments presented in the tables of these original works [3, 4, 7] may be useful in many cases, for most practical purposes in nuclear structure our calculators are no longer constrained by the size of these arguments and, e.g., calculations of coefficients involving nuclei populated to very high spin ($\gg 15\hbar$ cf. Ref. [3]) over a much wider range of multipolarities, magnetic substate projections, and for higher-rank tensors, may be readily evaluated. A further feature of this program allows users to regenerate the original tables of Refs. [3, 4, 7] in convenient machine-readable CSV and JSON formats. In particular, the defined vocabulary of the JSON schema provides implicit context regarding the nature and type of data held by the objects in their corresponding tables. These feature-rich formats, together with the associated contextual information described in this project, enable straightforward feature engineering allowing for mapping of raw data into feature vectors for further applications where labeled γ -ray angular distribution data may be useful. Extension of the tables to higher values of J , L , m , and k , as appropriate, can also be readily accomplished based on methods available within the PyGammaRAD library.

Given the essential role of angular momenta in these calculations, the software package is also loaded with a stand-alone general purpose vector-coupling calculator enabling exact analytical evaluation of Clebsch-Gordan, Racah, Wigner $3-j$, $6-j$, $9-j$, and Gaunt coefficients. These angular momentum coupling calculators have been verified against published data in various works on this topic [47, 44, 40, 58, 65, 60] and our results have also been checked using different proprietary [46] and open [49] tools. Notably, our vector-coupling methods can also accommodate and reproduce very large-integer couplings [47, 60]. This calculator combines the full apparatus of angular distribution calculations with a complete suite of vector-coupling algorithms that extends its use beyond the principal γ -ray angular-distribution focus to a broader range of quantum mechanical applications that involve the coupling of angular momenta, e.g., computation of molecular structure integrals based on Hartree-Fock-Roothaan methods [67]. Finally, the small modular nature of PyGammaRAD allows it to be integrated into larger codebases where calculations of angular distributions or angular momentum coupling coefficients are needed.

Acknowledgments

This work was performed under the auspices of the Department of Energy National Nuclear Security Administration by the Nuclear Science and Security Consortium under Award Number DE-NA0003996 at the University of California, Berkeley. Additional support was received through the Lawrence Berkeley National Laboratory under Contract No. DE-AC02-05CH11231 and by the Los Alamos National Laboratory under Contract No. 89233218CNA000001. We thank Dr. M. Cromaz for his valuable feedback and review of the manuscript. One of the authors, Aaron, thanks Mr. William Bryn Hurst for his belief that still carries me through.

Appendix A. Angular momentum coupling calculator

Although the development of a vector-coupling calculator is not a novel concept in itself, and indeed many such tools already exist that can satisfy this need, e.g., Refs. [40, 41, 42, 43, 44, 45, 58, 47, 59, 60, 49, 46], for convenience, the `PyGammaRAD` software package is also loaded with a complete angular momentum calculator that may serve a wide range of applications beyond the primary motivation of the work outlined in this paper.

The computational treatment of vector-coupling coefficients has a long history rooted in Fortran implementations. The foundational program of Tamura [40] provided subroutines for the Clebsch-Gordan and Racah coefficients based on direct evaluation of sum formulae; however, as noted by Wills [41], this approach is susceptible to numerical overflow and round-off errors for large angular momenta in single precision, owing to the rapid growth of the intermediate factorials. Wills [41] addressed this for the Clebsch-Gordan coefficient by recasting the summation as a nested product, thereby avoiding repeated exponentiation of logarithmic sums. This improvement was subsequently extended to the Wigner 6- j symbol by Bretz [42]. Srinivasa Rao and Venkatesh [43] then reformulated both the Clebsch-Gordan and Racah coefficients in terms of generalized hypergeometric functions of unit argument, specifically sets of ${}_3F_2(1)$ and ${}_4F_3(1)$ hypergeometric series, showing that this approach attains the same numerical accuracy as the Wills-Bretz method while achieving a further speed advantage of approximately 5–15%. A distinct approach was taken by Stone and Wood [44], who developed a Fortran package based on exact root-rational-fraction (RRF) arithmetic that addressed the limitations of large prime-factor storage which could cause integer overflow. Collectively, these Fortran implementations [40, 41, 42, 43, 44]

represent important milestones in the algorithmic development of vector-coupling coefficient computation, and their role in the lineage leading to modern implementations is recognized here. Computer algebra system approaches were subsequently developed by Lai [45], who wrote programs in the Maple language capable of computing both algebraic formulae and numerically exact values of Wigner $3-j$, $6-j$, and $9-j$ symbols; comparable symbolic capabilities are also available in Mathematica [46]. These tools, however, require proprietary software environments. Stevenson [47] later addressed this accessibility concern with a suite of Java applets, openly available and capable of evaluating Clebsch-Gordan, Wigner $3-j$, $6-j$, and $9-j$ symbols, and which remains accessible via GitHub [48]. More recently, the third-party library `SymPy` [49] addressed this need through symbolic manipulation and arbitrary-precision evaluation within the open Python ecosystem. `PyGammaRAD` represents a natural continuation of the open-source philosophy operating within the Python environment based on a conventional numerical computing approach, whilst broadening the scope to include the full angular distribution formalism. An important contribution to the efficient handling of these coefficients in large-scale computations was made by Rasch and Yu [59], who presented storage schemes for precalculated Wigner $3-j$, $6-j$, and Gaunt coefficients that exploit the full set of symmetries these symbols possess, achieving retrieval speeds more than an order of magnitude faster than recursion-based evaluation. A further important contribution in this area was developed by Johansson and Forssén [60] who demonstrated rapid precision evaluation of Wigner symbols using prime factorization and multiword integer arithmetic with an emphasis on couplings of very large angular momenta. Fast evaluation of Gaunt and Clebsch-Gordan coefficients was also demonstrated by Xu [68, 69] based on a forward substitution lower-triangle linear system recurrence algorithm. The `PyGammaRAD` implementation described here does not adopt a precalculation-and-storage strategy; instead, coefficients are evaluated on demand using the direct sum formulae described in the subsections below. This approach prioritises simplicity, portability, and exact arithmetic over raw retrieval speed, and is appropriate for the angular distribution applications motivating this work where the number of required coefficient evaluations is modest.

In the strict sense, it is only the Clebsch-Gordan and Racah coefficients that are needed to evaluate the angular distribution coefficients presented earlier in Sect 3.1. However, given their close relationship to the Wigner $3-j$, $6-j$, and $9-j$ symbols typically used to describe coupling and recoupling

schemes involving angular momentum in a quantum mechanical context, to broaden the utility of the `PyGammaRAD` software package, we provide methods to readily evaluate all aforementioned angular momentum coefficients and symbols. To that end, we also include a method for direct evaluation of the Gaunt coefficient since it can be decomposed into a generalization of Wigner 3- j symbols. The algorithms upon which these quantities are based are defined explicitly below.

Appendix A.1. Clebsch-Gordan coefficient

The Clebsch-Gordan coefficient describes the coupling of angular momentum vectors j_1 and j_2 , each with magnetic substate projections m_1 and m_2 , respectively, to form a system of total angular momentum j and corresponding projection m . This coupling scheme may be expressed as

$$\begin{aligned} \langle j_1 m_1 j_2 m_2 | j m \rangle &= \delta_{m, m_1 + m_2} \Delta(j_1 j_2 j) \sqrt{(j+m)!(j-m)!(2j+1)} \\ &\times \sqrt{(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!} \\ &\times \sum_{v=v_{\min}}^{v_{\max}} \left[\frac{1}{(j-j_2+m_1+v)!(j-j_1-m_2+v)!} \right. \\ &\times \left. \frac{(-1)^v}{v!(j_1+j_2-j-v)!(j_1-m_1-v)!(j_2+m_2-v)!} \right], \end{aligned} \quad (\text{A.1})$$

where the triangular (or delta) factor is given by

$$\Delta(j_1 j_2 j) = \sqrt{\frac{(j_1+j_2-j)!(j_1-j_2+j)!(-j_1+j_2+j)!}{(j_1+j_2+j+1)!}}. \quad (\text{A.2})$$

Only angular momentum vectors that satisfy the triangle inequalities condition result in the formation of a (j_1, j_2, j) triad arising from anti-parallel $(j_1 - j_2)$ and parallel $(j_1 + j_2)$ couplings:

$$|j_1 - j_2| \leq j \leq j_1 + j_2. \quad (\text{A.3})$$

The summation over v in Eq. (A.1) runs over integers such that

$$\begin{aligned} v_{\min} &= \max(0, \max(-(j-j_2+m_1), -(j-j_1-m_2))); \\ v_{\max} &= \min(j_1+j_2-j, \min(j_1-m_1, j_2+m_2)). \end{aligned} \quad (\text{A.4})$$

The phase convention for the Clebsch-Gordan coefficients evaluated using Eq. (A.1) follows that of Condon and Shortley [70]. Additional conditions also apply regarding the magnetic quantum number projections: If j_i is integral then so must be all corresponding m_i projections, i.e., if $2j_i \bmod 2 = 0$ then $2m_i \bmod 2 = 0$. Likewise, for half-integral j_i and m_i projections, if $2j_i \bmod 2 = 1$ then $2m_i \bmod 2 = 1$. Also, for both integral and half-integral cases, each projection must satisfy the relation $|m_i| \leq j_i$. Finally, the sum of the coupling projections must combine to give the total angular momentum projection, i.e., $m_1 + m_2 = m$.

Appendix A.2. Wigner 3- j symbol

The Wigner 3- j symbol is based on the same sum formula used in Eq. (A.1) to calculate the Clebsch-Gordan coefficient and the two quantities are related through the following expression

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m \end{pmatrix} = (-1)^{j+m+2j_1} \frac{1}{\sqrt{2j+1}} \langle j_1 - m_1 j_2 - m_2 | jm \rangle. \quad (\text{A.5})$$

The same conditions regarding the magnetic substate projections for the Clebsch-Gordan coefficient also apply here in the calculation of the Wigner 3- j symbol, as does the triangle rule [Eq. (A.3)] for triad couplings. Note that for projections m_1 , m_2 , and m given in a Wigner 3- j symbol, the projections must couple as $(-m_1) + (-m_2) = m$ (i.e., $m_1 + m_2 + m = 0$).

Appendix A.3. Gaunt coefficient

Gaunt coefficients arise in vector-coupling calculations where a triple product of spherical harmonics integrated over a solid angle is needed; this coefficient is included in PyGammaRAD for completeness. Using the notation $\langle l_2 m_2 | l_1 m_1 | l_3 m_3 \rangle = Y_{l_1 m_1 l_2 m_2}^{l_3 m_3}$ [65], we may then express the Gaunt coefficient as

$$\begin{aligned} Y_{l_1 m_1 l_2 m_2}^{l_3 m_3} &= \int_{\Omega=0}^{\Omega=4\pi} Y_{l_1, m_1}(\Omega) Y_{l_2, m_2}(\Omega) Y_{l_3, m_3}(\Omega) d\Omega \\ &= \int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} Y_{l_1, m_1}(\theta, \phi) Y_{l_2, m_2}(\theta, \phi) Y_{l_3, m_3}(\theta, \phi) \sin \theta d\theta d\phi, \end{aligned} \quad (\text{A.6})$$

where $Y_{l,m}(\Omega)$ is a spherical harmonic defined by orbital angular momentum and magnetic quantum numbers l and m , respectively, and $d\Omega = \sin\theta d\theta d\phi$ is the infinitesimal element of solid angle subtended. This Gaunt coefficient, as given by the integral of Eq. (A.6), can be conveniently calculated as a generalization of two Wigner 3- j symbols using the following expression

$$G(l_1 l_2 l_3, m_1 m_2 m_3) = \sqrt{\frac{(2l_1 + 1)(2l_2 + 1)(2l_3 + 1)}{4\pi}} \times \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \quad (\text{A.7})$$

Gaunt coefficients are governed by selection rules and have non-zero values only if $m_1 + m_2 + m_3 = 0$, the triangle condition $|l_1 - l_2| \leq l_3 \leq l_1 + l_2$ is met, and $(l_1 + l_2 + l_3) \bmod 2 = 0$.

Appendix A.4. Racah coefficient

The Racah coefficient $W(j_1 j_2 J j_3; J_{12} J_{23})$ involves the coupling of three angular momenta vectors. Here, there are two associated coupling schemes. In the first coupling scheme, j_1 couples with j_2 to form J_{12} , and then this resultant J_{12} vector couples with j_3 to form total angular momentum J , i.e., $|((j_1 j_2) J_{12}, j_3) J\rangle$. The other coupling scheme involves coupling j_2 with j_3 to give J_{23} , and j_1 next couples with J_{23} to give the same resultant total angular momentum J , i.e., $|((j_2 j_3) J_{23}, j_1) J\rangle$. Both coupling schemes result in complete orthonormal bases for the $(2j_1 + 1)(2j_2 + 1)(2j_3 + 1)$ dimensional space, and the scalar product of these two angular momentum bases yields the Racah recoupling coefficient. Explicitly, we may evaluate this coefficient as

$$W(j_1 j_2 J j_3; J_{12} J_{23}) = \Delta(j_1 j_2 J_{12}) \Delta(J j_3 J_{12}) \Delta(j_1 J J_{23}) \Delta(j_2 j_3 J_{23}) \times w(j_1 j_2 J j_3; J_{12} J_{23}), \quad (\text{A.8})$$

where each of the four triangular factors may be evaluated analogously to Eq. (A.2), subject to satisfaction of the triangle rule embodied by the inequality given in Eq. (A.3), and

$$w(j_1 j_2 J j_3; J_{12} J_{23}) \equiv \sum_{z=z_{\min}}^{z_{\max}} \left[\frac{(-1)^{z+\beta_1} (z+1)!}{(z-\alpha_1)!(z-\alpha_2)!(z-\alpha_3)!(z-\alpha_4)!} \times \frac{1}{(\beta_1 - z)!(\beta_2 - z)!(\beta_3 - z)!} \right], \quad (\text{A.9})$$

where

$$\begin{aligned}
\alpha_1 &= j_1 + j_2 + J_{12}; \\
\alpha_2 &= J + j_3 + J_{12}; \\
\alpha_3 &= j_1 + J + J_{23}; \\
\alpha_4 &= j_2 + j_3 + J_{23},
\end{aligned} \tag{A.10}$$

and

$$\begin{aligned}
\beta_1 &= j_1 + j_2 + J + j_3; \\
\beta_2 &= j_1 + j_3 + J_{12} + J_{23}; \\
\beta_3 &= j_2 + J + J_{12} + J_{23}.
\end{aligned} \tag{A.11}$$

The summation over z in Eq. (A.9) is finite and extends over limits given by

$$\begin{aligned}
z_{\min} &= \max(\alpha_1, \alpha_2, \alpha_3, \alpha_4); \\
z_{\max} &= \min(\beta_1, \beta_2, \beta_3).
\end{aligned} \tag{A.12}$$

Appendix A.5. Wigner 6- j symbol

Both the Racah coefficients and Wigner 6- j symbols are a generalization of Clebsch-Gordan coefficients and Wigner 3- j symbols that arise in the coupling of three angular momenta. The Wigner 6- j symbol and the Racah coefficient are related through a phase factor given by

$$\left\{ \begin{array}{ccc} j_1 & j_2 & J_{12} \\ j_3 & J & J_{23} \end{array} \right\} = (-1)^{j_1+j_2+j_3+J} W(j_1 j_2 J j_3; J_{12} J_{23}). \tag{A.13}$$

However, because of their higher number of symmetry properties, Wigner 6- j symbols are often preferred in many algorithms, e.g., Ref. [59], as a more efficient means of storing the recoupling coefficients². Clearly, the usual constraint of the triangle rule given by Eq. (A.3) also applies to angular momentum couplings involved in the evaluation of Wigner 6- j symbols.

²The 6- j symbol possesses a total of 144 symmetries, corresponding to the $3! \cdot 4!$ row and column permutations of an associated 3×4 Regge array [59], compared with the 24 tetrahedral symmetries more commonly cited in standard texts. Rasch and Yu [59] exploit this full set of symmetries to devise an efficient storage and retrieval scheme for precalculated 6- j symbols, which can be indexed via an ordered one-dimensional array without redundancy.

In our implementation, the `am_formulae.Racah` class serves as the computational engine for evaluation of both the Racah coefficient and the Wigner 6- j symbol. The `am_methods.AngularMomentumCalculations.racah` function, which evaluates $W(j_1 j_2 J j_3; J_{12} J_{23})$ directly through `racah(j1, j2, J, j3, J12, J23)`, calls the `am_formulae.Racah` class with arguments reordered to match the internal convention, specifically as `Racah(j1, j2, J12, j3, J, J23)`. The `am_methods.AngularMomentumCalculations.symb6j` function, which evaluates the Wigner 6- j symbol

$$\left\{ \begin{array}{ccc} j_1 & j_2 & J_{12} \\ j_3 & J & J_{23} \end{array} \right\},$$

passes its arguments directly to the `am_formulae.Racah` class as `Racah(j1, j2, J12, j3, J, J23)`, with no reordering (i.e., in the same order as the function `symb6j(j1, j2, J12, j3, J, J23)`), since the internal attribute ordering of the `am_formulae.Racah` class was chosen to match the row-by-row layout of the Wigner 6- j symbol. The relationship between the two calling conventions follows directly from Eq. (A.13): comparing the Wigner 6- j symbol layout with the Racah coefficient arguments $W(j_1 j_2 J j_3; J_{12} J_{23})$ shows that the top row of the Wigner 6- j array (j_1, j_2, J_{12}) corresponds to (j_1, j_2, J_{12}) of the Racah coefficient, while the bottom row (j_3, J, J_{23}) corresponds to (j_3, J, J_{23}) , confirming the positional rearrangement applied in the `racah` wrapper.

Appendix A.6. Wigner 9- j symbol

The Wigner 9- j symbols are a generalization of Clebsch-Gordan coefficients, Wigner 3- j , and Wigner 6- j symbols arising from coupling four angular momenta. As with the Racah coefficient (or Wigner 6- j symbol), the coupling of angular momenta can be achieved in different ways. In this case, we have $j_1 \otimes j_2 = J_{12}$ and $j_3 \otimes j_4 = J_{34}$, before coupling these resultants $J_{12} \otimes J_{34}$ to give total angular momentum J , i.e., $|((j_1, j_2)J_{12}, (j_3, j_4)J_{34})J\rangle$. In the alternate coupling scheme, $j_1 \otimes j_3 = J_{13}$ and $j_2 \otimes j_4 = J_{24}$, before coupling $J_{13} \otimes J_{24} = J$, i.e., $|((j_1, j_3)J_{13}, (j_2, j_4)J_{24})J\rangle$. Again, both coupling schemes provide a complete orthonormal basis for the $(2j_1+1)(2j_2+1)(2j_3+1)(2j_4+1)$ dimensional space and the scalar product of the associated functions for these coupling schemes gives the Wigner 9- j symbol.

For computational purposes and to help simplify the coding task, it is convenient to decompose the Wigner 9- j symbol into its triple product of

constituent Wigner 6- j symbols [71, 72]:

$$\begin{aligned} \left\{ \begin{array}{ccc} j_1 & j_2 & J_{12} \\ j_3 & j_4 & J_{34} \\ J_{13} & J_{24} & J \end{array} \right\} &= \sum_{x=x_{\min}}^{x_{\max}} (-1)^{2x} (2x+1) \\ &\times \left\{ \begin{array}{ccc} j_1 & j_2 & J_{12} \\ J_{34} & J & x \end{array} \right\} \left\{ \begin{array}{ccc} j_3 & j_4 & J_{34} \\ j_2 & x & J_{24} \end{array} \right\} \left\{ \begin{array}{ccc} J_{13} & J_{24} & J \\ x & j_1 & j_3 \end{array} \right\}. \end{aligned} \quad (\text{A.14})$$

The summation in Eq. (A.14) extends over all values of x that satisfy the triangle inequality theorem in the corresponding triangular factors, such that no factorial has a negative argument within the limits given by

$$\begin{aligned} x_{\min} &= \max(|j_1 - J|, |j_2 - J_{34}|, |j_3 - J_{24}|); \\ x_{\max} &= \min(j_1 + J, j_2 + J_{34}, j_3 + J_{24}). \end{aligned} \quad (\text{A.15})$$

Due to its symmetry properties, any Wigner 9- j symbol is invariant under reflection through its diagonals. Therefore, the Wigner 9- j symbol in Eq. (A.14) has different equivalent representations, for example

$$\begin{aligned} \left\{ \begin{array}{ccc} j_1 & j_3 & J_{13} \\ j_2 & j_4 & J_{24} \\ J_{12} & J_{34} & J \end{array} \right\} &= \sum_{x=x_{\min}}^{x_{\max}} (-1)^{2x} (2x+1) \\ &\times \left\{ \begin{array}{ccc} j_1 & j_3 & J_{13} \\ J_{24} & J & x \end{array} \right\} \left\{ \begin{array}{ccc} j_2 & j_4 & J_{24} \\ j_3 & x & J_{34} \end{array} \right\} \left\{ \begin{array}{ccc} J_{12} & J_{34} & J \\ x & j_1 & j_2 \end{array} \right\}, \end{aligned} \quad (\text{A.16})$$

whereupon the limits of x are the same as those given in Eq. (A.15).

Computational methods based on formulations of alternative algorithms, rather than a summation over a triple product of Wigner 6- j symbols, have also been proposed elsewhere. Lai [45] demonstrated symbolic and numerically exact computation of Wigner 9- j symbols within the Maple computer algebra system, generating closed algebraic formulae for arbitrary angular momentum arguments. Wei [58] derived an algebraic expression for the Wigner 9- j symbol as a twofold summation of products of binomial coefficients, with a recursive algorithm for evaluating those binomials that avoids the computation of factorials of integers entirely, thereby eliminating the primary source of overflow in large-angular-momentum calculations.

The approach adopted in PyGammaRAD uses the triple Wigner 6- j product decomposition of Eq. (A.14), evaluated via the Racah sum formulae of the `am_formulae.Racah` class [Eqs. (A.8)–(A.12)]. In this implementation, the

summation loop variable is stored internally as $h = 2x$, an integer. This ensures that the loop has a step size of two and always increments over integers rather than half-integers, with the phase factor $(-1)^{2x} \equiv (-1)^h$ and the degeneracy weight $(2x + 1) \equiv (h + 1)$ obtained directly from h . This allows us to recast Eq. (A.14) into the formal expression upon which the PyGammaRAD algorithm for the Wigner 9- j symbol is computed:

$$\left\{ \begin{array}{ccc} j_1 & j_2 & J_{12} \\ j_3 & j_4 & J_{34} \\ J_{13} & J_{24} & J \end{array} \right\} = \sum_{h=h_{\min}}^{h_{\max}} (-1)^h (h + 1) \quad (\text{A.17})$$

$$\times \left\{ \begin{array}{ccc} j_1 & j_2 & J_{12} \\ J_{34} & J & \frac{h}{2} \end{array} \right\} \left\{ \begin{array}{ccc} j_3 & j_4 & J_{34} \\ j_2 & \frac{h}{2} & J_{24} \end{array} \right\} \left\{ \begin{array}{ccc} J_{13} & J_{24} & J \\ \frac{h}{2} & j_1 & j_3 \end{array} \right\},$$

with associated summation bounds given by

$$\begin{aligned} h_{\min} &= h_{\max} \bmod 2; \\ h_{\max} &= 2 \min(j_1 + J, j_2 + J_{34}, j_3 + J_{24}). \end{aligned} \quad (\text{A.18})$$

The lower bound $h_{\min}$ in Eq. (A.18) is set such that h shares the same parity as $h_{\max}$, ensuring h increments in steps of two starting from either $h_{\min} = 0$ (integral) or $h_{\min} = 1$ (half-integral). It is worth noting that the phase factor in Eq. (A.17) will always be positive since $(-1)^h = 1$ regardless of whether h takes even or odd parity. Each of the three Wigner 6- j symbols in Eq. (A.17) is evaluated by passing arguments directly to the `am_formulae.Racah` class in the row-by-row layout as described in Appendix A.5, with h representing the intermediate angular momentum.

References

- [1] N. Bohr, Neutron Capture and Nuclear Constitution, *Nature* (London) 137 (1936) 344. doi:10.1038/137344a0.
- [2] A. E. Litherland, A. J. Ferguson, Gamma-ray angular correlations from aligned nuclei produced by nuclear reactions, *Can. J. Phys.* 39 (1961) 788. doi:10.1139/p61-089.
- [3] T. Yamazaki, Tables of Coefficients for Angular Distribution of Gamma Rays from Aligned Nuclei, *Nucl. Data Sect. A* 3 (1967) 1. doi:10.1016/S0550-306X(67)80002-8.

- [4] E. Der Mateosian, A.W. Sunyar, Tables of Attenuation Coefficients for Angular Distributions of Gamma Rays from Partially Aligned Nuclei, *At. Data and Nucl. Data Tables* 13 (1974) 391. doi:10.1016/0092-640X(74)90007-2.
- [5] L. C. Biedenharn, M. E. Rose, Theory of Angular Correlation of Nuclear Radiations, *Rev. Mod. Phys.* 25 (1953) 729–777. doi:10.1103/RevModPhys.25.729.
- [6] D. R. Hamilton, On Directional Correlation of Successive Quanta, *Phys. Rev.* 58 (1940) 122–131. doi:10.1103/PhysRev.58.122.
- [7] H.J. Rose, D.M. Brink, Angular Distributions of Gamma Rays in Terms of Phase-Defined Reduced Matrix Elements, *Rev. Mod. Phys.* 39 (1967) 306. doi:10.1103/RevModPhys.39.306.
- [8] K. S. Krane, R. M. Steffen, R. M. Wheeler, Directional correlations of gamma radiations emitted from nuclear states oriented by nuclear reactions or cryogenic methods, *Nucl. Data Tables* 11 (1973) 351–406. doi:10.1016/S0092-640X(73)80016-6.
- [9] K. Krane, Solid-angle correction factors for coaxial Ge(Li) detectors, *Nucl. Instrum. Methods* 98 (1972) 205. doi:10.1016/0029-554X(72)90099-7.
- [10] A.M. Hurst, SolidAngles, <https://github.com/AaronMHurst/SolidAngles>, GitHub repository.
- [11] H. Frauenfelder, Angular Correlation of Nuclear Radiation, *Annu. Rev. Nucl. Sci.* 2 (1953) 129–162. doi:10.1146/annurev.ns.02.120153.001021.
- [12] J. J. Smith, et al., Gamma–gamma angular correlation analysis techniques with the GRIFFIN spectrometer, *Nucl. Instrum. Methods Phys. Res. A* 922 (2019) 47. doi:10.1016/j.nima.2018.10.097.
- [13] C. Morse, et al., The polarization sensitivity of GRETINA, *Nucl. Instrum. Methods Phys. Res. Sect. A* 1025 (2022) 166155. doi:10.1016/j.nima.2021.166155.

- [14] U. Fano, Description of States in Quantum Mechanics by Density Matrix and Operator Techniques, *Rev. Mod. Phys.* 29 (1957) 74. doi:10.1103/RevModPhys.29.74.
- [15] G. Racah, Theory of Complex Spectra. II, *Phys. Rev.* 62 (1942) 438. doi:10.1103/PhysRev.62.438.
- [16] S. Ali, et al., Exhibition of multiple band structures of different kind in ^{142}Eu , *Phys. Rev. C* 111 (2025) 024303. doi:10.1103/PhysRevC.111.024303.
- [17] A.N. Deacon, et al., Single-particle and collective structures in ^{55}Cr and ^{55}V , *Phys. Rev. C* 83 (2011) 064305. doi:10.1103/PhysRevC.83.064305.
- [18] D. Steppenbeck, et al., Magnetic rotation and quasicollective structures in ^{58}Fe : Influence of the $\nu g_{9/2}$ orbital, *Phys. Rev. C* 85 (2012) 044316. doi:10.1103/PhysRevC.85.044316.
- [19] M. Piiparinen, et al., High-spin spectroscopy of the ^{142}Eu , ^{143}Eu and ^{144}Eu nuclei, *Nucl. Phys. A* 605 (1996) 191. doi:10.1016/0375-9474(96)00157-1.
- [20] D. Rudolph, et al., Rotational and single-particle structures in ^{83}Zr , *Z. Phys. A* 338 (1991) 139. doi:10.1007/BF01284787.
- [21] M. Cromaz, et al., High spin spectroscopy of ^{167}Hf , *Phys. Rev. C* 59 (1999) 2406. doi:10.1103/PhysRevC.59.2406.
- [22] C.J. Chiara, et al., Spectroscopy in the $Z = 49$ $^{108,110}\text{In}$ isotopes: Lifetime measurements in shears bands, *Phys. Rev. C* 64 (2001) 05314. doi:10.1103/PhysRevC.64.05314.
- [23] L.A. Bernstein, et al., Onset of collectivity in neutron deficient $^{196,198}\text{Po}$, *Phys. Rev. C* 52 (1995) 621. doi:10.1103/PhysRevC.52.621.
- [24] Y. Zheng, et al., γ -ray linear polarization measurements and $(g_{9/2})^{-3}$ neutron alignment in ^{91}Ru , *Phys. Rev. C* 87 (2013) 044328. doi:10.1103/PhysRevC.87.044328.
- [25] B.F. Lv, et al., Tilted precession bands in ^{135}Nd , *Phys. Rev. C* 103 (2021) 044308. doi:10.1103/PhysRevC.103.044308.

- [26] N. Buorn, et al., Magnetic properties of the $\nu j_{15/2}$ intruder orbital in the superdeformed minimum of ^{197}Pb , Eur. Phys. J. A 9 (2000) 29. doi:10.1007/s100500070052.
- [27] D.R. Jensen, et al., Evidence for Second-Phonon Nuclear Wobbling, Phys. Rev. Lett. 89 (2002) 142503. doi:10.1103/PhysRevLett.89.142503.
- [28] D.R. LaFosse, et al., Collective structures and band termination in ^{107}Sb , Phys. Rev. C 62 (2000) 014305. doi:10.1103/PhysRevC.62.014305.
- [29] E.S. Paul, et al., γ -ray spectroscopy of neutron-deficient ^{110}Te . I. Low- and intermediate-spin structures, and II. High-spin smooth-terminating structures, Phys. Rev. C 76 (2007) 034322, 034323, parts I and II; DOIs: 10.1103/PhysRevC.76.034322, 10.1103/PhysRevC.76.034323.
- [30] R.M. Clark, et al., Yrast and near-yrast excitations up to high spin in $^{100}_{48}\text{Cd}_{52}$, Phys. Rev. C 61 (2000) 044311. doi:10.1103/PhysRevC.61.044311.
- [31] J.F. Smith, et al., Identification of excited states in ^{117}Cs : Systematics of the $\nu(h_{11/2})^2$ alignment, Phys. Rev. C 63 (2001) 024319. doi:10.1103/PhysRevC.63.024319.
- [32] S. Zhu, et al., Level structure of the neutron-rich $^{56,58,60}\text{Cr}$ isotopes: Single-particle and collective aspects, Phys. Rev. C 74 (2006) 064315. doi:10.1103/PhysRevC.74.064315.
- [33] R. M. Diamond, E. Matthias, J. O. Newton, F. S. Stephens, Nuclear Alignment in Heavy-Ion Reactions, Phys. Rev. Lett. 16 (1966) 1205. doi:10.1103/PhysRevLett.16.1205.
- [34] J. O. Newton, F. S. Stephens, R. M. Diamond, K. K. Kotajima, E. Matthias, Angular Distributions of Gamma Rays Produced in Reactions with Heavy Ions, Nucl. Phys. A 95 (1967) 357. doi:10.1016/0375-9474(67)90507-6.
- [35] A.M. Hurst, L.A. Bernstein, T. Kawano, A.M. Lewis, K. Song, The Baghdad Atlas: A relational database of inelastic neutron-scattering ($n, n'\gamma$) data, Nucl. Instrum. Methods Phys. Res. Sect. A 995 (2021) 165095. doi:10.1016/j.nima.2021.165095.

- [36] L. Miriot-Jaubert, et al., First Study of the Pygmy Dipole Resonance via Neutron Inelastic Scattering at GANIL-SPIRAL2/NFS: Benchmarking the Method on the ^{12}C 2^+ Excited State, *Acta Phys. Pol. B* 18 (2025) 2–A35. doi:10.5506/APhysPolBSupp.18.2-A35.
- [37] T. Lauritsen, et al., γ -ray angular correlations, distributions and linear polarization in tracking arrays, *Eur. Phys. J. A* 61 (2025) 76. doi:10.1140/epja/s10050-025-01533-5.
- [38] B. Longfellow, et al., In-beam angular distribution and linear polarization measurements with GRETINA using a simple energy-ordering approach, *Nucl. Instrum. Methods Phys. Res. Sect. A* 1082 (2026) 170932. doi:10.1016/j.nima.2025.170932.
- [39] A. Chalil, O. Litaize, T. Materna, A. Chebboubi, Quantification of spin alignment in fission by simultaneous treatment of γ and conversion electron angular distributions, *Phys. Rev. C* 109 (2024) 014605. doi:10.1103/PhysRevC.109.014605.
- [40] T. Tamura, Angular Momentum Coupling Coefficients, *Comp. Phys. Comm.* 1 (1970) 337, Program title: GEOMETRICAL COEFFICIENT; Catalogue number: ABMA; Erratum: *Angular Momentum Coupling Coefficients* (Correction Deck), *Comp. Phys. Comm.* 2, 173–174 (1971). doi:10.1016/0010-4655(70)90034-2.
- [41] J.G. Wills, On the evaluation of angular momentum coupling coefficients, *Comp. Phys. Comm.* 2 (1971) 381. doi:10.1016/0010-4655(71)90030-0.
- [42] V. Bretz, An improved method for calculation of angular momentum coupling coefficients, *Acta Physica* 40 (1976) 255. doi:10.1007/BF03157502.
- [43] K. Srinivasa Rao, K. Venkatesh, New fortran programs for angular momentum coefficients, *Comp. Phys. Comm.* 15 (1978) 227, Program title: ANGMOM; Catalogue number: ACYK. doi:10.1016/0010-4655(78)90093-0.
- [44] A.J. Stone, C.P. Wood, Root-rational-fraction package for exact calculation of vector-coupling coefficients, *Comp. Phys. Comm.* 21 (1980)

- 195, Program title: Root Rational Fraction Package; Catalogue number: ABVN; <https://www-stone.ch.cam.ac.uk/wigner.shtml>. doi: 10.1016/0010-4655(80)90040-5.
- [45] S.-T. Lai, Computation of Algebraic Formulas for Wigner 3- j , 6- j , and 9- j Symbols by Maple, Int. J. Quantum Chem. 52 (1994) 593. doi: 10.1002/qua.560520303.
 - [46] Wolfram Research, Inc., Mathematica, Version 14.3, <https://www.wolfram.com/mathematica>, Champaign, IL (2025).
 - [47] P.D. Stevenson, Analytic angular momentum coupling coefficient calculators, Comp. Phys. Comm. 147 (2002) 853, Program title: Cleblet; Catalogue number: ADQG. doi:10.1016/S0010-4655(02)00462-9.
 - [48] P.D. Stevenson, Clebsch-o-Matic, <https://github.com/pdstevenson/clebsch-o-matic>, GitHub repository.
 - [49] A. Meurer, et al., SymPy: symbolic computing in Python, PeerJ Comp. Sci. 3 (2017) e103. doi:10.7717/peerj-cs.103.
 - [50] E. S. Macias, W. D. Ruhter, D. C. Camp, R. G. Lanier, A program for calculating gamma-gamma directional correlation coefficients and mixing ratios, Comp. Phys. Comm. 11 (1976) 75, Program title: ANGCOR; Catalogue number: ABPG. doi:[https://doi.org/10.1016/0010-4655\(76\)90041-2](https://doi.org/10.1016/0010-4655(76)90041-2).
 - [51] R.J. Rouse, G.L. Strubble, R.G. Lanier, L.G. Mann, A program for calculating gamma-gamma directional correlation coefficients and angular distribution coefficients for gamma rays of mixed multipolarities from partially aligned nuclei, Comp. Phys. Comm. 15 (1978) 107, Program title: THDST; Catalogue number: ABMZ. doi:10.1016/0010-4655(78)90085-1.
 - [52] A.M. Hurst, PyGammaRAD: Python project for Gamma-Ray Angular Distributions, <https://github.com/AaronMHurst/PyGammaRAD>, GitHub repository.
 - [53] A.M. Hurst, PyGammaRAD: Python project for Gamma-Ray Angular Distributions, <https://pypi.org/project/PyGammaRAD/>, Python Package Index (PyPI) repository.

- [54] D.M. Brink, G.R. Satchler, Angular Momentum, 2nd Edition, Clarendon Press, Oxford, 1968.
- [55] V. Devanathan, Angular Momentum Techniques in Quantum Mechanics, Kluwer Academic Publishers, New York, Boston, Dordrecht, London, Moscow, 2002.
- [56] A.R. Edmonds, Angular Momentum in Quantum Mechanics, Princeton University Press, Princeton, New Jersey, 1957.
- [57] M.E. Rose, Elementary Theory of Angular Momentum, John Wiley & Sons, Inc., New York, London, 1957.
- [58] L. Wei, New formula for 9- j symbols and their direct calculation, Comput. Phys. 12 (1998) 632. doi:10.1063/1.168745.
- [59] J. Rasch, A.C.H. Yu, Efficient Storage Scheme for Precalculated Wigner $3j$, $6j$ and Gaunt Coefficients, SIAM J. Sci. Comput. 25 (2003) 1416. doi:10.1137/S1064827503422932.
- [60] H.T. Johansson, C. Forssén, Fast and Accurate Evaluation of Wigner $3j$, $6j$, and $9j$ Symbols Using Prime Factorization and Multiword Integer Arithmetic, SIAM J. Sci. Comput. 38 (2016) A376. doi:10.1137/15M1021908.
- [61] K. S. Krane, $E2$, $M1$ multipole mixing ratios in even-even nuclei, $A \geq 152$, At. Data Nucl. Data Tables 16 (1975) 383. doi:10.1016/0092-640X(75)90018-2.
- [62] K. S. Krane, R. M. Steffen, Determination of the $E2/M1$ Multipole Mixing Ratios of the Gamma Transitions in ^{110}Cd , Phys. Rev. C 2 (1970) 724. doi:10.1103/PhysRevC.2.724.
- [63] K. Alder, R. M. Steffen, The Electromagnetic Interaction in Nuclear Spectroscopy, North-Holland, Amsterdam, 1975.
- [64] K.S. Krane, Tables of Coefficients for Analysis of Angular Distribution of Gamma Radiation from Oriented Nuclei, Tech. Rep. LA-4677, Los Alamos National Laboratory (June 1971).

- [65] S.A. Yükcü, S. Atalay, N. Yükcü, E. Öztekin, Calculation of the Gaunt coefficients over real spherical harmonics, *Can. J. Phys.* 103 (2025) 321. doi:10.1139/cjp-2024-0161.
- [66] T.J. Ypma, Historical Development of the Newton-Raphson Method, *SIAM Rev.* 37 (1995) 531. doi:10.1137/1037125.
- [67] C.C.J. Roothaan, A Study of Two-Center Integrals Useful in Calculations on Molecular Structure. I, *J. Chem. Phys.* 19 (1951) 1445. doi:10.1063/1.1748100.
- [68] Y.-L. Xu, Fast evaluation of the Gaunt coefficients, *Math. Comput.* 65 (1996) 1601.
- [69] Y.-L. Xu, Fast evaluation of the Gaunt coefficients: recursive approach, *J. Comput. Appl. Math.* 85 (1997) 53. doi:https://doi.org/10.1016/S0377-0427(97)00128-3.
- [70] E.U. Condon, G.H. Shortley, *Theory of Atomic Spectra*, Cambridge University Press, 1935.
- [71] A. Messiah, *Quantum Mechanics*, Vol. 2, Amsterdam, Netherlands: North-Holland, 1962.
- [72] B.W. Shore, D.H. Menzel, *Principles of Atomic Spectra*, John Wiley & Sons, New York, 1968.